A Problem with Presidents

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Introduction

This report finds the longest and shorted lived presidents based on how many days they lived which can be seen in the tables below. The methodology used includes both a table for longest lived presidents, including and excluding current living presidents. The only different between the two is the inclusion of Joe Biden in the longest-lived list. It then finds the statistical summary of the data then plots the data into a histogram.

```
#imports all of the data
presidents <- read.csv("presidential_data.csv")
president_name <- presidents$name
birth_date <- mdy(presidents$birth)
death_date <- mdy(presidents$death)

#creates the needed data
year_of_birth <- year(birth_date)
lived_days <- as.integer(death_date - birth_date)
lived_months <- lived_days/30.44
lived_years <- year(death_date) - year(birth_date)

#finds the longest and shortest lived presidents
df <- data.frame(president_name, lived_days)
longest <- df[order(-df$lived_days),][1:10,]
# longest
shortest <- df[order(df$lived_days),][1:10,]
# shortest
```

```
#repeats for current living presidents
death_date_l <- replace(death_date, is.na(death_date), Sys.Date())
lived_days_l <- as.integer(death_date_l - birth_date)
df_l <- data.frame(president_name, lived_days_l)
longest_l <- df_l[order(-df_l$lived_days_l),][1:10,]
shortest_l <- df_l[order(df_l$lived_days_l),][1:10,]
```

```
#makes a table of the longest living presidents
longest |>
  gt() |>
  tab_header(
    title = md("**Longest Lived Presidents**"),
    subtitle = md("(Excludes Living)")
  ) |>

  cols_label(
    president_name = "President",
    lived_days = "Days Lived"
  )|>
  tab_options(
    heading.title.font.size = "large",
    heading.subtitle.font.size = "small",
    latex.use_longtable = TRUE
  )
```

Longest Lived Presidents

(Excludes Living)

President	Days Lived
Jimmy Carter	36614
George Bush	34504
Gerald Ford	34133
Ronald Reagan	34088
John Adams	33119
Herbert Hoover	32943
Harry S. Truman	32373
James Madison	31150
Thomas Jefferson	30397
Richard Nixon	29688

```
longest_l |>
  gt() |>
  tab_header(
    title = md("**Longest Lived Presidents**"),
    subtitle = md("(Includes Living)")
  ) |>

  cols_label(
    president_name = "President",
    lived_days_l = "Days Lived"
  )|>
  tab_options(
    heading.title.font.size = "large",
    heading.subtitle.font.size = "small",
    latex.use_longtable = TRUE
  )
```

Longest Lived Presidents
(Includes Living)

President	Days Lived
Jimmy Carter	36614
George Bush	34504
Gerald Ford	34133
Ronald Reagan	34088
John Adams	33119
Herbert Hoover	32943
Harry S. Truman	32373
James Madison	31150
Joe Biden	30660
Thomas Jefferson	30397

```
shortest |>
  gt() |>
  tab_header(
    title = md("**Shortest Lived Presidents**"),
    # subtitle = md("(Excludes Living)")
  ) |>

  cols_label(
    president_name = "President",
    lived_days = "Days Lived"
  )|>
  tab_options(
    heading.title.font.size = "large",
    latex.use_longtable = TRUE
  )
```

Shortest Lived Presidents

President	Days Lived
John F. Kennedy	16978
James A. Garfield	18202
James K. Polk	19583
Abraham Lincoln	20516
Chester Arthur	20863
Warren G. Harding	21091
William McKinley	21412
Theodore Roosevelt	21985
Calvin Coolidge	22099
Franklin Roosevelt	23082

```
# shortest_l |>
#   gt() |>
```

```
#   tab_header(
#     title = md("**Shortest Lived Presidents**"),
#     subtitle = md("(Includes Living)")
#   ) |>
#
#   cols_label(
#     president_name = "President",
#     lived_days = "Days Lived"
#   )
```

```r
#finds the statistical summaries of the data
#there is no mode and weighted average does not make sense in this context so they have been excluded
mean_lived = mean(lived_days, na.rm = TRUE)
median_lived = median(lived_days, na.rm = TRUE)
# mode_lived = Mode(lived_days, na.rm = TRUE)
max_lived = max(lived_days, na.rm = TRUE)
min_lived = min(lived_days, na.rm = TRUE)
sd_lived = SD(lived_days, na.rm = TRUE)

#makesa data frame of the data
summary <- data.frame(stats = c("Mean",
                                "Median",
                                "Maximum",
                                "Minimum",
                                "Standard Deviation"),
                      vals = c( mean_lived,
                                median_lived,
                                max_lived,
                                min_lived,
                                sd_lived))

#creates a table of the statistic sums
summary |> gt() |>
  tab_header(
    title = md("**President Statistics**"),
    subtitle = md("(Based on Lived Days)")
  )|>
  cols_label(
    stats = "Statistic",
    vals = "Value"
  )|>
  tab_options(
    heading.title.font.size = "large",
    latex.use_longtable = TRUE
  )
```

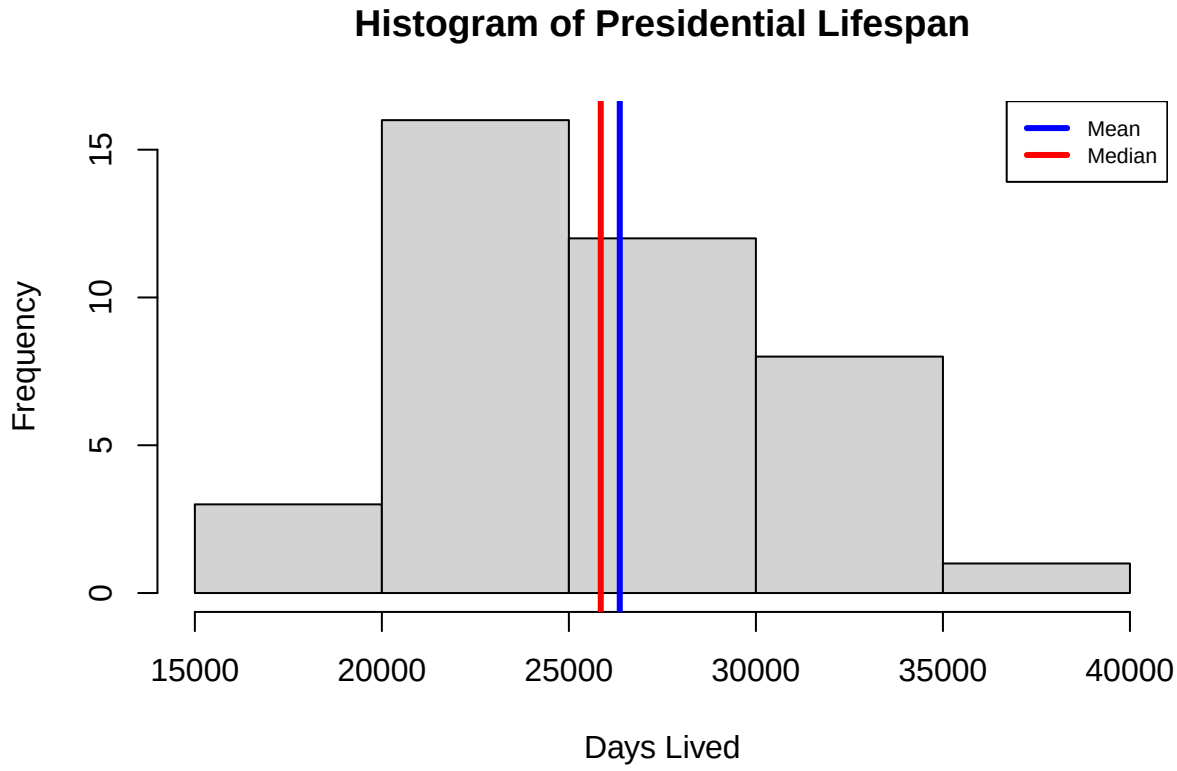
President Statistics
(Based on Lived Days)

Statistic	Value
Mean	26360.100
Median	25851.500
Maximum	36614.000
Minimum	16978.000
Standard Deviation	4831.908

```r
#makes a histogram
hist(lived_days[!is.na(lived_days)],
      main = "Histogram of Presidential Lifespan",
      xlab = "Days Lived")
```

```r
#plots the mean and the median
abline(v = mean_lived,   col = "blue",  lwd = 3)
abline(v = median_lived,   col = "red",  lwd = 3)

#makes a legend of the data
legend("topright",
       legend = c("Mean", "Median"),
       col = c("blue", "red"),
       lwd = 3,
       cex = 0.7)
```



Conclusion

Based on the conclusions found in the report, the longest lived president is Jimmy Carter and the shortest is John F. Kennedy. The data also appears to the skewed to the right based on both the histogram as well as the mean being higher than the median.